

MATH 415-01 Advanced Linear Algebra
Homework #2
Assigned: January 30, 2026
Due: February 6, 2026

A few notes.

1. Matrices are your friend. You are not allowed to take a random object and suddenly create a matrix out of it. For example, polynomials don't just become traditional vectors. Start with definitions, and if a system of equations comes out of that definition, then feel free to use a matrix to solve the system.
 2. You may use technology to row reduce matrices. Solving systems of equations by hand takes a lot of work. Row reduction is your friend, and the **RREF** command on your calculator is your friend, as well.
 3. Knowing how to interpret the presence of pivots in the rows and columns of RREF matrices is very valuable.
 4. Know the difference between the definition of a basis and The Basis Theorem.
 5. You are adults. You can staple/fasten your homework before you turn it in.
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1. Let

$$\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}.$$

- (a) Compute $\hat{\mathbf{v}}$.
 - (b) Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? Justify your conclusion.
 - (c) Compute the distance between $\mathbf{u}$ and $\mathbf{v}$.
 - (d) Compute the angle between $\mathbf{u}$ and $\mathbf{v}$ in radians.
2. Verify the *parallelogram law* for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,
- $$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2.$$
3. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. Prove that if the distance from $\mathbf{u}$ to $\mathbf{v}$ is equal to the distance from $\mathbf{u}$ to $-\mathbf{v}$, then $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.
4. Let W be a subspace of $\mathbb{R}^n$. Prove that $W^\perp$ is a subspace of $\mathbb{R}^n$.
5. Consider

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}.$$

- (a) Show that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ form a basis of $\mathbb{R}^3$ by checking orthogonality. Justify this approach by stating the appropriate theorem.
 - (b) Express
- $$\mathbf{x} = \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$$
- as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. Do not solve a system of equations as part of your calculation.
6. Let $\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ -7 \end{bmatrix}$. Write $\mathbf{x}$ as the sum of two orthogonal vectors, one in $\text{Span}\{\mathbf{v}\}$ and one orthogonal to $\mathbf{v}$. Show all of your work.